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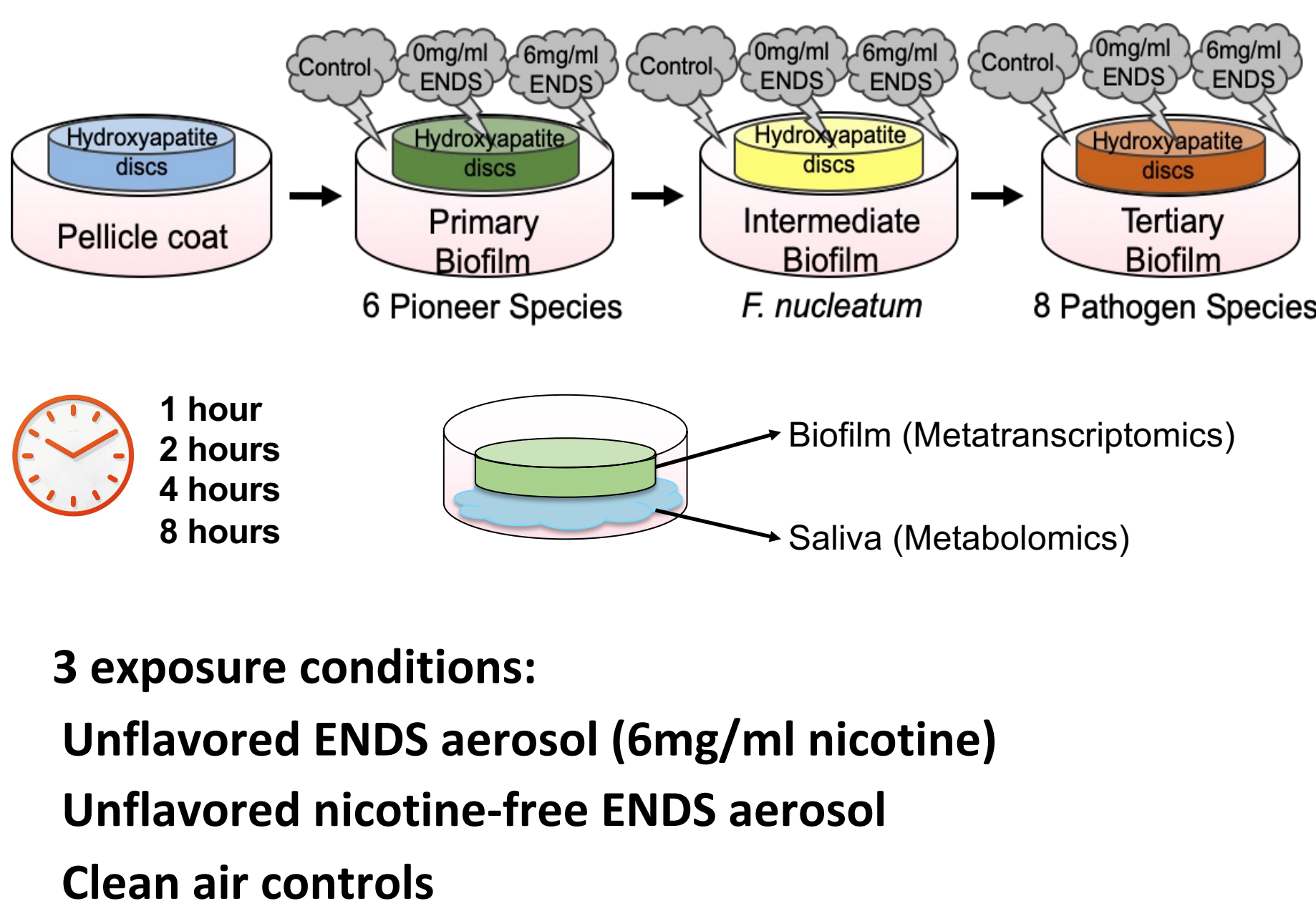
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Abstract: E-cigarettes have emerged as an exposomal factor of great concern to human health. We aimed to test the hypothesis that e-cigarette vapor is metabolized in the oral cavity by the indigenous microbiome, leading to structural and functional alterations in oral biofilms. We combined *in vitro* models of commensal-rich and pathogen-rich microcosm communities with untargeted metabolomics, metatranscriptomics, and fluorescent microscopy, and verified the results in human samples. Spectral deconvolution of 4,215 peaks identified 969 exposomal and endogenous metabolites that mapped to 23 metabolic pathways. Vapor characteristics and biofilm composition affected metabolite profiles. Metabolites generated by commensal-rich biofilms contained antimicrobial, anti-fungal and anti-bacterial compounds, while pathogen-rich biofilms metabolized nicotine-containing vapor using the pyridine and pyrrolidine pathways. Both communities generated endogenous metabolites that mapped to quorum sensing functions. Several metabolites were also verified in the saliva of current, never, and former smokers who vape. Xenobiotic degradation, capsule, peptidoglycan, and glycosaminoglycan biosynthesis were upregulated in commensal-rich communities, while genes encoding organic carbon-compound metabolism, antimicrobial resistance and secretion systems were over-expressed in pathogen-rich biofilms. Topographical analysis revealed an architecture characterized by low surface-area to biovolume ratio, high biomass, and diffusion distance only in commensal-rich biofilms. In conclusion, our data suggest that bacterial metabolism of e-cigarette vapor triggers a quorum-sensing-regulated stress response which mediates the formation of dense, exopolysaccharide-rich biofilms in health-compatible communities and antibiotic resistance and virulence amplification in disease-associated communities. These findings provide a biological rationale for emergent evidence that vapers have a higher incidence of dental caries, gingival inflammation, and antimicrobial resistance.

Background

- E-cigarettes have surged in popularity as an alternative to traditional tobacco products, with an estimated 41 million adults and 3.6 million adolescents in the United States reported to have used e-cigarettes in 2020.
- Concerns have arisen regarding the health implications of e-cigarette vapor exposure, containing a mixture of chemicals including nicotine and flavorants, particularly concerning its direct impact on the oral cavity during inhalation.
- The oral microbiome, comprising a diverse array of bacteria, fungi, and other microorganisms, plays a crucial role in maintaining oral health, with alterations linked to various oral diseases.
- Advanced analytical techniques such as untargeted metabolomics and metatranscriptomics are employed to comprehensively analyze the metabolic and transcriptional responses of oral biofilms to e-cigarette vapor exposure.
- Integration of findings from *in vitro* models with validation in human samples is conducted to understand the complex interactions between e-cigarette vapor and the oral microbiome, shedding light on its potential implications for oral health outcomes.

Methodology



Schematic representation of the methodology and the downstream methods used in order to carry out this study.

Results

Figure 1: Effect of Nicotine Concentration on Metabolite Diversity and Generation in E-cigarette Vapor Exposure

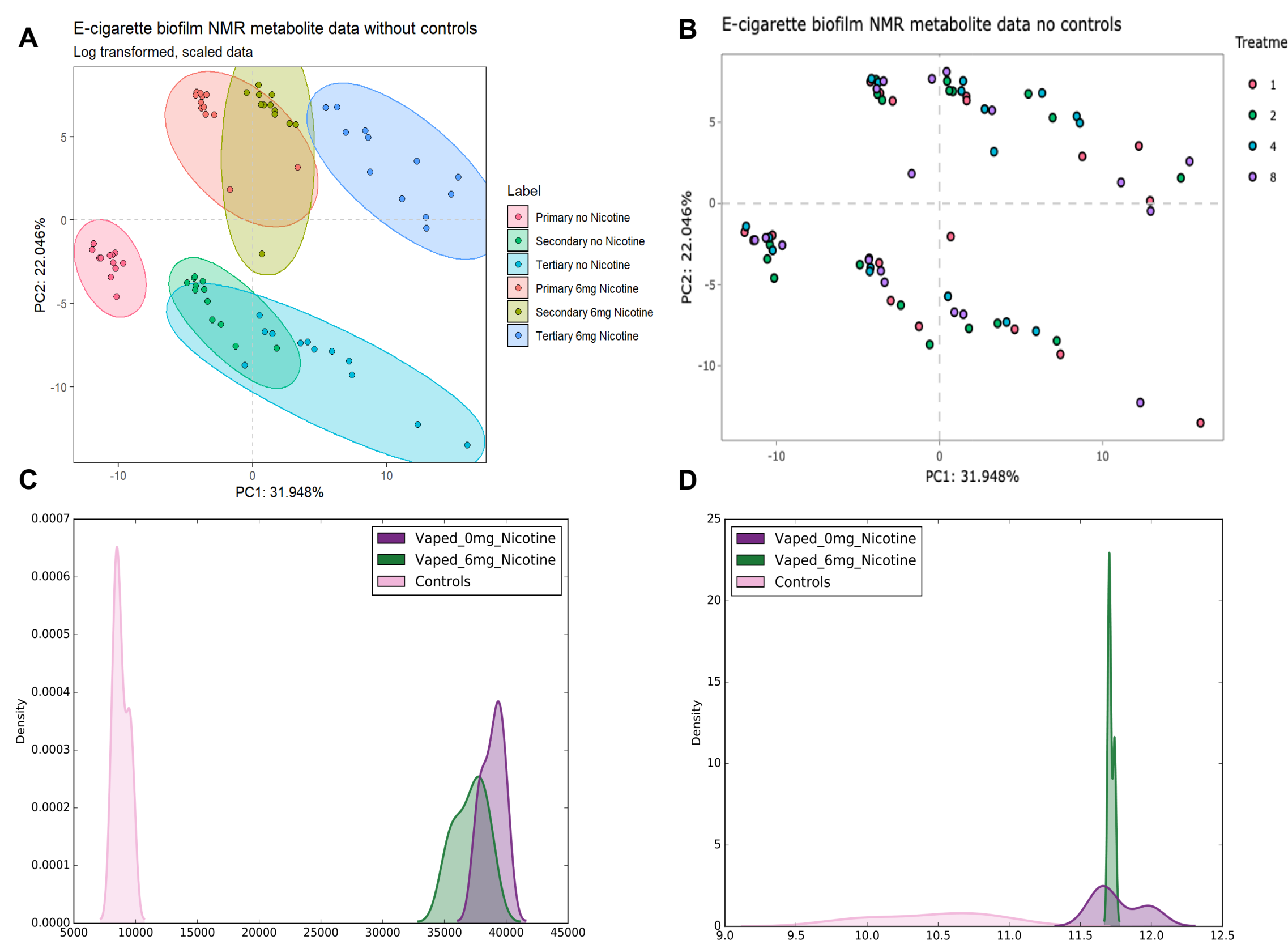


Figure 1: Metabolite diversity and generation over time. PCA revealed nicotine concentration (0mg versus 6mg) as a source of variation along with biofilm diversity (F1A). The majority of the metabolites were generated within one hour of exposure to e-cigarette vapor (1B). Chao1 (C) and Shannon (D) indices demonstrated the greatest level of metabolite diversity when biofilms were exposed to nicotine-containing vapor.

Figure 2: Both microbial composition and biofilm composition drive metabolomic profile

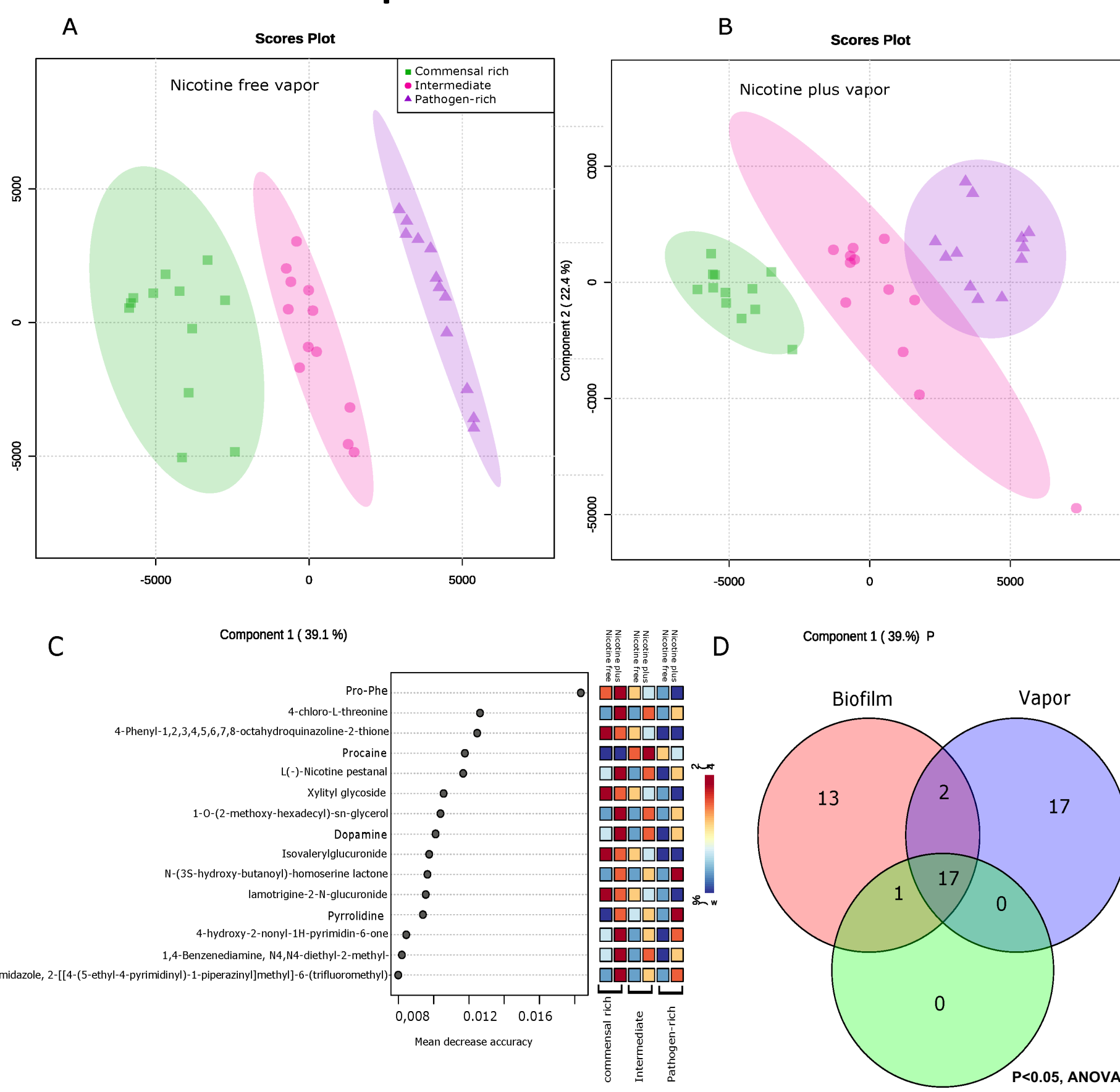


Figure 2: Partial least squares discriminant analysis (PLSDA) demonstrated that 39.1% of the variation between biofilms was explained by component 1 following exposure to nicotine-free ENDS, while exposure to nicotine-containing vapor accounted for 37.6% of the variation of component 1 (2A and 2B). A machine-learning algorithm trained on the metabolite profiles identified 15 compounds as discriminants of biofilm composition and vapor type (2C). Covariate analysis demonstrated a significant interaction between biofilm diversity and vapor composition ($p < 0.05$, ANOVA) (2D).

Figure 3: Impact of E-cigarette Vapor Exposure on Pathway Enrichment Across Commensal, Intermediate, and Pathogen-Rich Biofilms

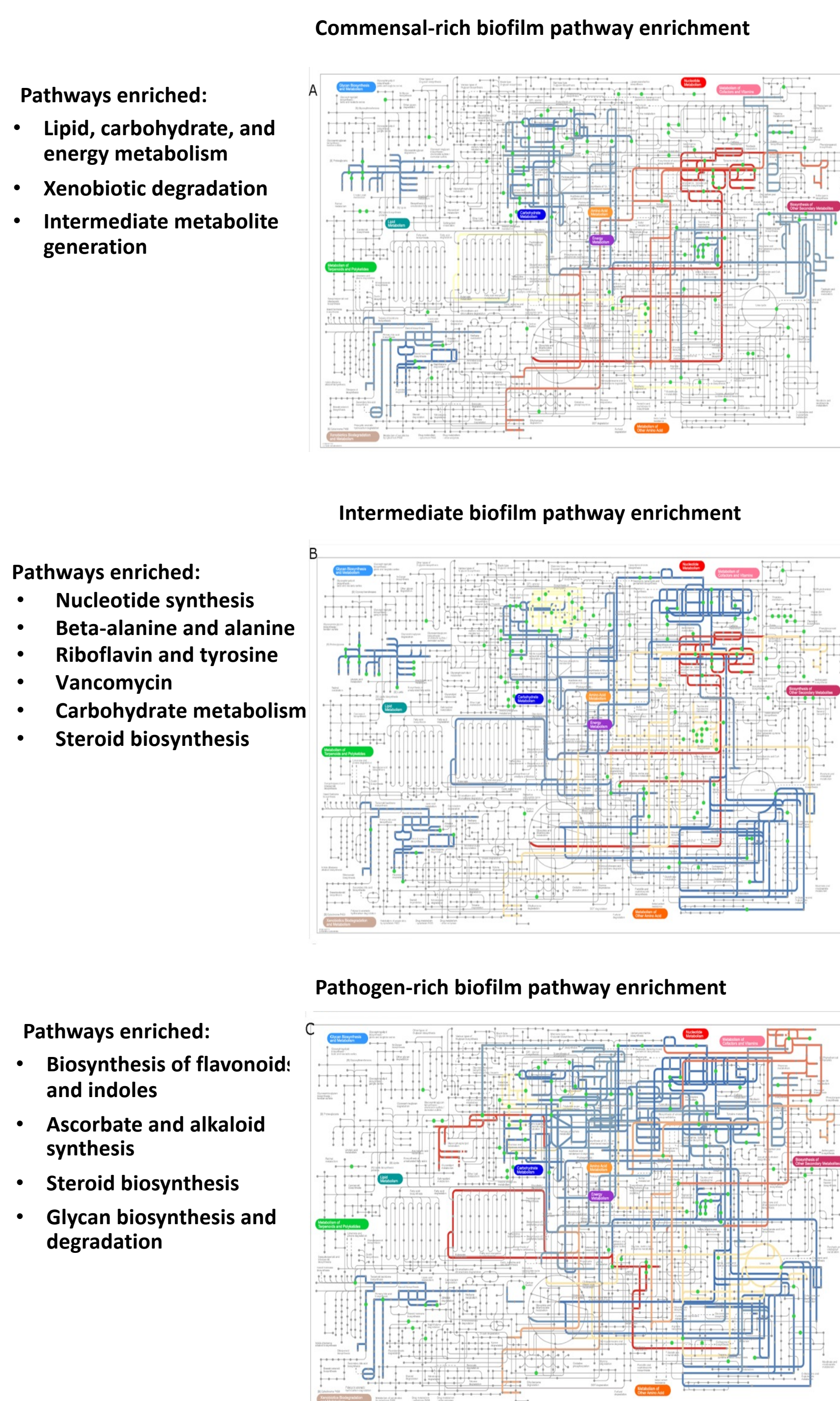


Figure 3: Pathway enrichment analysis identifies key biological functions impacted by vapor exposure. Figure 3A demonstrates pathways that are upregulated (blue lines) and down-regulated (red lines) in commensal-rich communities following exposure to e-cigarette vapor (3A). Figure 3B demonstrates these pathways in intermediate biofilms and Figure 3C in pathogen-rich biofilms. Data from nicotine-free and nicotine-plus vapor were combined for this analysis, since there were few differences between groups at the pathway level. Grey lines indicate that no pathway was matched and green dots represent matched compounds.

Figure 4: E-cigarette aerosol induces quorum-sensing regulated gene expression

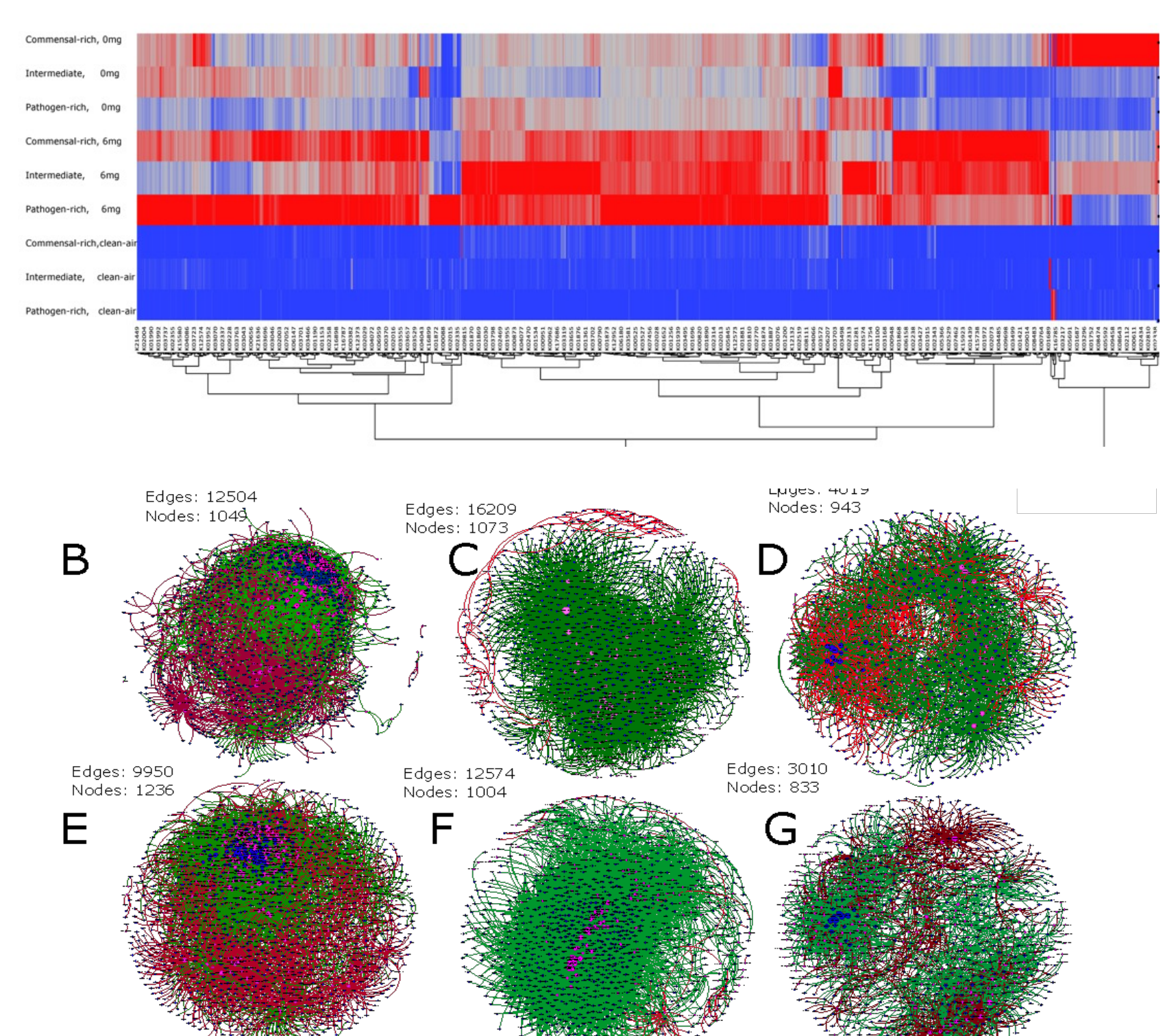


Figure 4: E-cigarettes induce quorum-sensing regulated gene expression in oral biofilms. Heatmap of KEGG orthologs that were upregulated following exposure to nicotine-free or nicotine plus vapor or clean-air is shown in Figure 4A. Co-occurrence networks between metabolites and microbial transcripts in each group are shown in 4B to 4G. Commensal-rich biofilms are shown in (B and E), Intermediate in (4C and 4F), and pathogen-rich in (4D and 4G). Nicotine-free vapor exposure is represented in Figures B-D and nicotine-plus vapor exposure is shown in Figures 4E-4G. Each network graph contains nodes (circles) and edges (lines). Nodes represent metabolites (pink) and KEGG-annotated transcripts (blue), and edges represent Spearman's rho. Edges are colored green for positive correlation and red for negative correlation. Only significant correlations ($P < 0.05$, t test) with a coefficient of at least 0.80 are shown.

Figure 5: E-cigarette exposure alters biofilm topography

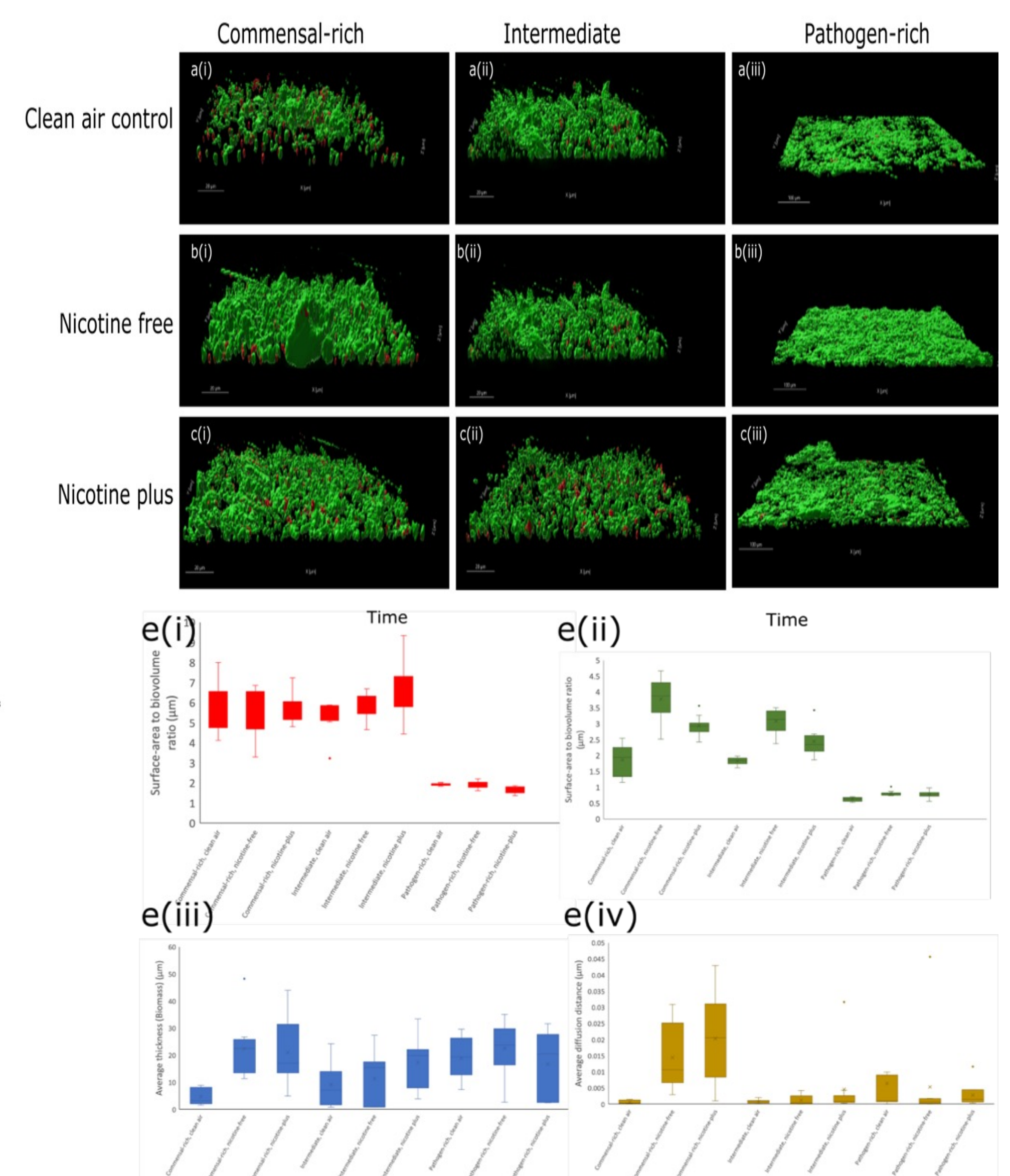


Figure 5: Confocal images of commensal-rich, intermediate, and pathogen-rich biofilms exposed to nicotine-free or nicotine-containing e-cigarette vapor, along with clean air controls (5a-c). Changes in biofilm area, surface area to biovolume ratio, biomass, and diffusion distance over 8 hours depicted (5e). Significant differences indicated by shared symbols ($P < 0.001$, Dunn's test).

Conclusion

- Oral biofilms are capable of metabolizing e-cigarette aerosol mostly occurs within first hour of exposure.
- Bacterial transcription is significantly altered in an e-cigarette rich environment in addition nicotine further alters transcription.
- Annotated metabolites and transcriptional pathways point to antibacterial agents playing a key role in why biofilms are impacted by e-cigarette aerosol.
- Biofilm structure is significantly altered, likely via quorum sensing and cell wall, capsule, and ECM synthesis.